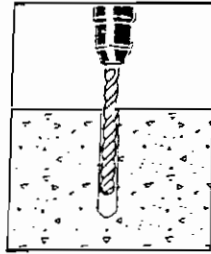


**ATTACHMENT 8 – MANUFACTURER LITERATURE FROM POWER’S
FASTENERS, INC. ON INSTALLATION REQUIREMENTS FOR THE
ADHESIVE ANCHORING SYSTEM**
(4 pages)

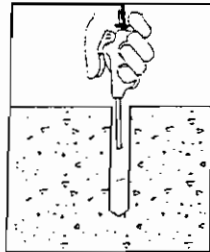
52.8 Power-Fast™ Installation Guidelines

52.8.1 Threaded Rods or Reinforcing Bars in Solid Base Materials

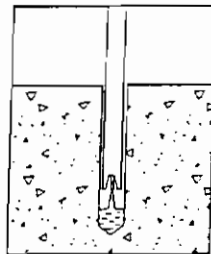
Drill a hole to the size and embedment required. The tolerances of the drill bit used should meet the requirements of ANSI Standard B212.15.



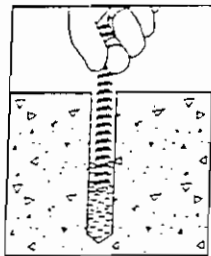
Blow the hole clean with compressed air, brush the hole, and blow it clean again. Holes should be clean and sound. They may be dry or damp, but should be free of standing water or frost. If using reinforcing bars, be sure the bar will fit into the drilled hole. If a larger hole is required, the diameter should be as close as possible to the diameter of the reinforcing bar.



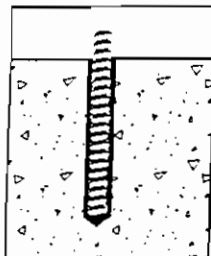
Be sure to properly balance the mixing nozzle prior to dispensing and when changing cartridges or sausages. Fill the hole approximately half way with epoxy starting from the rear of the hole. Slowly withdraw the static mixing nozzle as the hole fills to avoid creating air pockets within the hole.



Push the threaded rod or reinforcing bar into the hole while turning slightly to insure positive dispersion of the epoxy. Be sure the rod is fully seated at the bottom of the hole and that some epoxy has flowed from the top of the hole. The threaded rod or reinforcing bar used should be free of dirt, grease, oil, or other foreign material.

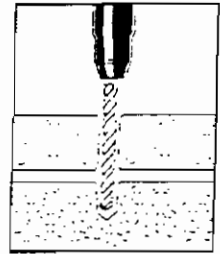


Allow the epoxy to cure for the specified time prior to applying any load. Do not disturb or load the anchor until it is fully cured.

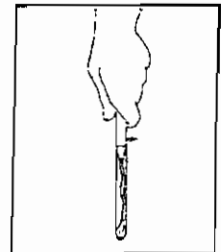


52.8.2 Threaded Rods in Hollow Base Materials

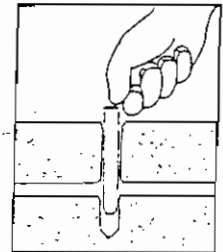
Drill a hole to the size and embedment for the required screen size. The tolerances of the drill bit used should meet the requirements of ANSI B212.15. Blow the hole clean with compressed air, brush the hole, and blow it clean again.



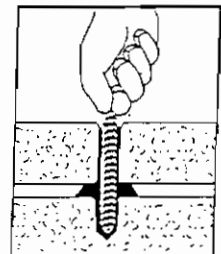
Properly balance the mixing nozzle prior to dispensing and when changing cartridges or sausages. Fill the screen tube with Power-Fast epoxy starting from the rear of the screen. The screen tube should be filled prior to inserting it in the hole.



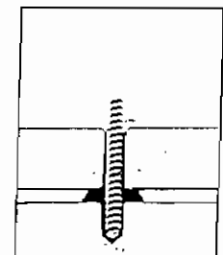
Insert the filled screen tube into the anchor hole until it is fully seated at the required embedment.



Push the threaded rod into the screen while turning slightly to insure positive dispersion of the epoxy. Be sure the rod is fully inserted to the end of the screen tube. The threaded rod or reinforcing bar used should be free of dirt, grease, oil, or other foreign material.



Allow the epoxy to cure for the specified time prior to applying any load. Do not disturb or load the anchor until it is fully cured.



Prior to use, read the box and carton label, Design Manual, Material Data Safety Sheet, and injection tool instructions.

52.9.2 Allowable Load Capacities for Installations Using Threaded Anchor Rod in Concrete

The bond strengths in the following table are based on applying a safety factor of 4 to the ultimate load capacities of the epoxy injection gel for three concrete compressive strengths. The allowable steel strength values are based on the AISC Manual of Steel Construction (Ninth Edition). Values for B-7 steel rod are based on A325 capacities. The allowable load used should be the lesser of the bond or steel strength.

Allowable Tension

Anchor Size (inches)	Drill Dia. (inches)	Embed. Depth (inches)	Allowable Bond Strength (lbs.)			Allowable Steel Strength (lbs.)				
			2,000 psi Concrete	4,000 psi Concrete	6,000 psi Concrete	A36 Steel	A307 Steel	B7 Steel	304 SS Steel	316 SS Steel
1/4	5/16	1	350	400	550	940	980	2,160	1,210	1,210
1/4	5/16	2	590	720	1,105	940	980	2,160	1,210	1,210
1/4	5/16	3	965	1,120	1,560	940	980	2,160	1,210	1,210
3/8	7/16	1-1/2	755	1,195	1,390	2,100	2,200	4,840	2,720	2,720
3/8	7/16	2-1/4	1,130	1,675	1,960	2,100	2,200	4,840	2,720	2,720
3/8	7/16	3-3/8	1,945	2,270	2,815	2,100	2,200	4,840	2,720	2,720
3/8	7/16	4-1/2	2,335	2,835	3,420	2,100	2,200	4,840	2,720	2,720
1/2	9/16	2	1,280	1,970	2,190	3,745	3,920	8,625	4,850	4,850
1/2	9/16	3	1,920	2,840	3,160	3,745	3,920	8,625	4,850	4,850
1/2	9/16	4-1/2	2,880	4,360	4,615	3,745	3,920	8,625	4,850	4,850
1/2	9/16	6	3,840	4,925	5,815	3,745	3,920	8,625	4,850	4,850
5/8	3/4	2-1/2	2,130	2,490	3,225	5,865	6,140	13,510	7,560	7,560
5/8	3/4	3-3/4	3,195	3,960	4,840	5,865	6,140	13,510	7,560	7,560
5/8	3/4	5-5/8	4,795	7,545	8,010	5,865	6,140	13,510	7,560	7,560
5/8	3/4	7-1/2	6,390	7,900	9,680	5,865	6,140	13,510	7,560	7,560
3/4	7/8	3	2,480	2,585	3,730	8,440	8,840	19,450	10,940	10,940
3/4	7/8	4-1/2	4,465	4,640	6,170	8,440	8,840	19,450	10,940	10,940
3/4	7/8	6-3/4	6,700	9,735	10,150	8,440	8,840	19,450	10,940	10,940
3/4	7/8	9	8,930	11,120	13,035	8,440	8,840	19,450	10,940	10,940
7/8	1	3-1/2	3,965	4,730	5,585	11,480	12,020	26,445	14,875	14,875
7/8	1	5-1/4	5,940	8,505	9,005	11,480	12,020	26,445	14,875	14,875
7/8	1	7-7/8	8,915	10,595	13,010	11,480	12,020	26,445	14,875	14,875
7/8	1	10-1/2	11,880	15,890	17,975	11,480	12,020	26,445	14,875	14,875
1	1-1/8	4	5,115	6,375	7,750	14,995	15,700	34,540	19,430	19,430
1	1-1/8	6	7,660	9,630	10,640	14,995	15,700	34,540	19,430	19,430
1	1-1/8	9	11,505	14,890	16,860	14,995	15,700	34,540	19,430	19,430
1	1-1/8	12	15,340	20,150	23,245	14,995	15,700	34,540	19,430	19,430
1-1/4	1-3/8	5	7,750	10,195	11,745	23,400	24,540	53,990	30,370	30,370
1-1/4	1-3/8	7-1/2	11,630	15,295	17,620	23,400	24,540	53,990	30,370	30,370
1-1/4	1-3/8	11-1/4	17,445	22,940	26,430	23,400	24,540	53,990	30,370	30,370
1-1/4	1-3/8	15	23,225	30,590	35,240	23,400	24,540	53,990	30,370	30,370
1-3/8	1-1/2	5-1/2	9,275	12,200	14,055	28,400	29,700	65,340	36,730	36,730
1-3/8	1-1/2	8-1/4	13,915	18,300	21,080	28,400	29,700	65,340	36,730	36,730
1-3/8	1-1/2	12-3/8	20,870	27,450	31,620	28,400	29,700	65,340	36,730	36,730
1-3/8	1-1/2	16-1/2	27,825	36,600	42,160	28,400	29,700	65,340	36,730	36,730
1-1/2	1-5/8	6	10,395	14,385	16,565	33,700	35,340	77,750	43,735	43,735
1-1/2	1-5/8	9	16,400	21,575	24,850	33,700	35,340	77,750	43,735	43,735
1-1/2	1-5/8	13-1/2	24,605	32,360	37,275	33,700	35,340	77,750	43,735	43,735
1-1/2	1-5/8	18	32,805	43,145	45,705	33,700	35,340	77,750	43,735	43,735

Allowable Shear

Anchor Size (inches)	Drill Dia. (inches)	Embed. Depth (inches)	Allowable Bond Strength (lbs.)			Allowable Steel Strength (lbs.)				
			2,000 psi Concrete	4,000 psi Concrete	6,000 psi Concrete	A36 Steel	A307 Steel	B7 Steel	304 SS Steel	316 SS Steel
1/4	5/16	1	620	620	620	485	490	1,030	625	625
3/8	7/16	1-1/2	1,620	1,620	1,620	1,090	1,100	2,310	1,400	1,400
1/2	9/16	2	2,780	2,780	2,780	1,940	1,960	4,115	2,500	2,500
5/8	3/4	2-1/2	4,415	4,415	4,415	3,040	3,070	6,450	3,915	3,915
3/4	7/8	3	7,315	7,315	7,315	4,380	4,420	9,280	5,635	5,635
7/8	1	3-1/2	9,015	9,015	9,015	5,950	6,010	12,620	7,665	7,665
1	1-1/8	4	13,285	13,285	13,285	7,770	7,850	16,485	10,010	10,010
1-1/4	1-3/8	5	20,865	20,865	20,865	12,150	12,270	25,770	15,645	15,645
1-3/8	1-1/2	5-1/2	24,365	24,365	24,365	14,690	14,850	31,160	18,920	18,920
1-1/2	1-5/8	6	28,195	28,195	28,195	17,490	17,670	37,110	22,530	22,530

52.9 Performance Data

52.9.1 Ultimate Load Capacities for Installations Using Threaded Anchor Rod in Concrete

The following load capacities are the ultimate or failure load for the epoxy based on tests using high strength anchor rod. The capacity of the anchor rod itself should be taken into account in the design of any anchorage. The allowable load capacities for the epoxy and five types of anchor rod are described in the next section.

Anchor Size (inches)	Drill Dia. (inches)	Embed. Depth (inches)	Torque Range (ft.-lbs.)	2,000 psi Concrete		4,000 psi Concrete		6,000 psi Concrete	
				Tension (lbs.)	Shear (lbs.)	Tension (lbs.)	Shear (lbs.)	Tension (lbs.)	Shear (lbs.)
1/4	5/16	1	5 - 10	1,400	2,465	1,600	2,465	2,210	2,465
1/4	5/16	2	5 - 10	2,370	2,465	2,950	2,465	4,420	2,465
1/4	5/16	3	5 - 10	3,860	2,465	4,470	2,465	6,240	2,465
3/8	7/16	1-1/2	10 - 20	3,010	6,480	4,780	6,480	5,560	6,480
3/8	7/16	2-1/4	10 - 20	4,515	6,480	6,695	6,480	7,840	6,480
3/8	7/16	3-3/8	10 - 20	7,770	6,480	9,085	6,480	11,260	6,480
3/8	7/16	4-1/2	10 - 20	9,330	6,480	11,340	6,480	13,680	6,480
1/2	9/16	2	20 - 40	5,120	11,120	7,885	11,120	8,755	11,120
1/2	9/16	3	20 - 40	7,680	11,120	11,365	11,120	12,630	11,120
1/2	9/16	4-1/2	20 - 40	11,515	11,120	17,435	11,120	18,460	11,120
1/2	9/16	6	20 - 40	15,535	11,120	19,700	11,120	23,265	11,120
5/8	3/4	2-1/2	50 - 90	8,520	17,650	9,960	17,650	12,910	17,650
5/8	3/4	3-3/4	50 - 90	12,780	17,650	15,845	17,650	19,360	17,650
5/8	3/4	5-5/8	50 - 90	19,170	17,650	30,175	17,650	32,040	17,650
5/8	3/4	7-1/2	50 - 90	25,560	17,650	31,605	17,650	38,720	17,650
3/4	7/8	3	100 - 160	9,910	29,250	10,345	29,250	14,290	29,250
3/4	7/8	4-1/2	100 - 160	17,860	29,250	18,550	29,250	24,670	29,250
3/4	7/8	6-3/4	100 - 160	26,790	29,250	38,935	29,250	40,590	29,250
3/4	7/8	9	100 - 160	35,720	29,250	44,475	29,250	52,130	29,250
7/8	1	3-1/2	180 - 200	15,860	36,065	18,930	36,065	22,330	36,065
7/8	1	5-1/4	180 - 200	23,765	36,065	34,010	36,065	36,010	36,065
7/8	1	7-7/8	180 - 200	35,650	36,065	42,385	36,065	52,030	36,065
7/8	1	10-1/2	180 - 200	47,530	36,065	63,560	36,065	71,890	36,065
1	1-1/8	4	250 - 300	20,455	53,135	25,495	53,135	30,990	53,135
1	1-1/8	6	250 - 300	30,630	53,135	38,510	53,135	42,550	53,135
1	1-1/8	9	250 - 300	46,025	53,135	59,555	53,135	67,430	53,135
1	1-1/8	12	250 - 300	61,365	53,135	80,605	53,135	92,975	53,135
1-1/4	1-3/8	5	350 - 600	31,010	83,450	40,785	83,450	46,980	83,450
1-1/4	1-3/8	7-1/2	350 - 600	46,510	83,450	61,180	83,450	70,475	83,450
1-1/4	1-3/8	11-1/4	350 - 600	69,770	83,450	91,765	83,450	105,710	83,450
1-1/4	1-3/8	15	350 - 600	93,025	83,450	122,355	83,450	140,950	83,450
1-3/8	1-1/2	5-1/2	350 - 600	37,100	97,460	48,800	97,460	56,215	97,460
1-3/8	1-1/2	8-1/4	350 - 600	55,650	97,460	73,200	97,460	84,320	97,460
1-3/8	1-1/2	12-3/8	350 - 600	83,480	97,460	109,800	97,460	126,480	97,460
1-3/8	1-1/2	16-1/2	350 - 600	111,300	97,460	146,395	97,460	168,640	97,460
1-1/2	1-5/8	6	350 - 600	43,740	112,780	57,530	112,780	66,270	112,780
1-1/2	1-5/8	9	350 - 600	65,605	112,780	86,295	112,780	99,405	112,780
1-1/2	1-5/8	13-1/2	350 - 600	98,410	112,780	129,440	112,780	149,105	112,780
1-1/2	1-5/8	18	350 - 600	131,215	112,780	172,585	112,780	182,810	112,780